



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (paper-3)

– Hira Kumar

C++ Programming

- C++ = High level language & General purpose language (किसी भी काम के लिए)
- Application / Used in = System Software, Video game, Embedded systems, IOT device, AI application,...etc
- C++ Extension of C language. OR C++ is a super set of C language.
- Old name of C++ language is “C with Classes”, after 1983 its called C++.
- Designed / Developed by = Bjarne Stroustrup
- Year = 1979 (Invented) at AT&T Bell laboratories.
- Based on = Modular Programming (s/w की सबसे छोटी unit – module)
- Approach / follow – Bottom-UP Approach
- C++ → Platform independent source code
- Extension of C++ is = .cpp, .cxx, .cc (.C only for Unix/Linux OS)
- First Java Compiler & JVM (Java Virtual Machine) is written in C++
- C++ → C++ is one of the most popular languages for Competitive Programming because of STL (Standard template library) and high performance.
- C++ world 5th most used programming language in 2026.
(1. Python, 2. Javascript, 3. Java, 4. C++, 5. C#(C Sharp))

- C++ used Concept of Classes and object.
- C++ supports the OOP's = 4 pillar
 1. Abstraction (Hides implementation details and shows only essential features.)
 2. Polymorphism (One interface, many forms.)
 3. Inheritance (Acquiring properties and behaviour of one class into another class.e.g. – parent-child)
 4. Encapsulation (Wrapping data and methods into a single unit (Class))
- C++ used = Compiler (GCC(g++), MinGW, clang,MSVC)
- C++ called Earlier (before) = C with classes / C with Objects
- C++ is a case sensitive language.
- Used Compiler/Source code where C++ Run i.e. IDE (Integrated Development environment)
 - ➔ Turbo C++ / Borland C++
 - ➔ Code :: Block
 - ➔ Dev c++ (Developer C++)
 - ➔ Eclipse
 - ➔ Visual Studio Code (Microsoft)
 - GCC (GNU Compiler Collection)

• What is IDE ?

➔ Integrated Development Enviroment (एक तरह का tool box हैं, जिसमें debug,compiler,loder,linker,.....होते हैं)

IDE Consist of-

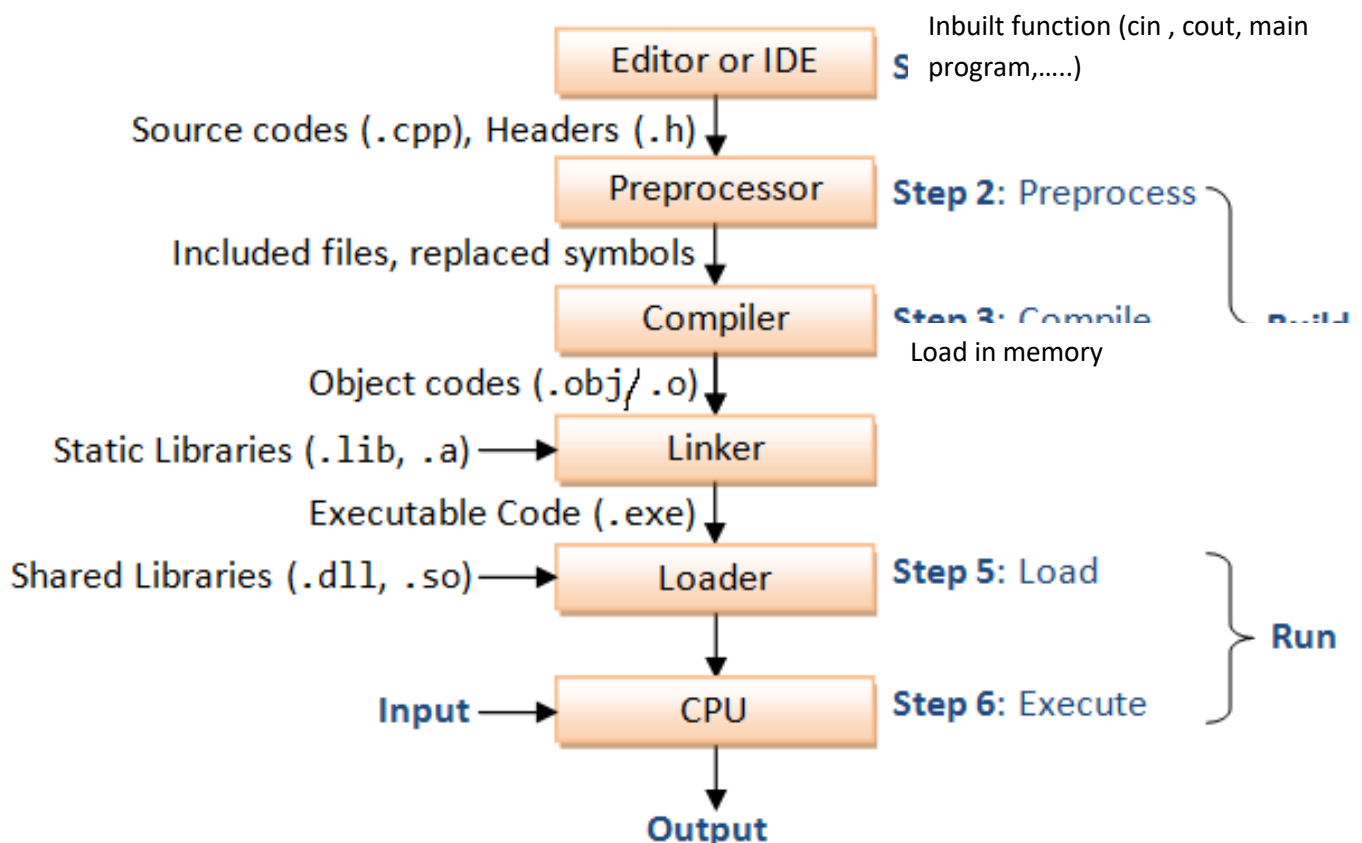
- i. Source Code Editor
- ii. Compiler
- iii. Linker

- iv. Debugger
- v. Build Tool

Difference Between C and C++

C	C++
1. C is Procedural <u>Language</u> .	1. C++ is non Procedural i.e Object oriented Language.
2. No virtual Functions are present in C	2. The concept of virtual Functions are used in C++.
3. In C, Polymorphism is not possible.	3. The concept of polymorphism is used in C++. Polymorphism is the most Important Feature of OOPS.
4. Operator overloading is not possible in C.	4. Operator overloading is one of the greatest Feature of C++.
5. Top down approach is used in Program <u>Design</u> .	5. Bottom up approach adopted in Program Design.
6. No namespace Feature is present in C Language.	6. Namespace Feature is present in C++ for <u>avoiding</u> Name collision.

❖ C++ Flow Diagram



Step 1: Preprocessing / Preprocessor

The first step in the compilation process is preprocessing. This step is where the compiler processes the source code and performs various tasks such as **including header files, expanding macros, and removing comments**. The result of this step is a modified version of the source code, which is then passed on to the next step of the compilation process.

Step 2: Compilation (Compiler)

The next step in the compilation process is compilation. This step is where the preprocessed **source code is compiled into object code(.obj / .o) or Machine code**. Object code is a low-level representation of the code that can be executed by a computer.

During the compilation step, the compiler reads the preprocessed source code and performs the following tasks: **Syntax analysis** (grammatically correct), **Code generation** (generates object code from the source code), **Semantic analysis** (checking for variable declarations, function definitions, and other language-specific features). If any error then show **Compiler error**.

Step 3: Linking / Linker

The final step in the compilation process is linking. This step is where the object files are linked together to form an executable program. The linker performs the following tasks: Symbol resolution, Relocation, and Output generation.

Object file + required libraries = Executable/binary file (.exe file → Windows)

Step 4: Optimization

After the executable file is generated, it can be further optimized to improve its performance. Optimization is the process of improving the efficiency of the compiled code by making it execute faster, use less memory, or both. Optimization can be done at various levels, including the compiler, linker, and operating system.

Step 5: Debugging

The Next step in the C++ compilation process is debugging. Debugging is the process of identifying and fixing errors in the compiled code. Debugging can be done using various tools, including debuggers, profilers, and memory checkers.

Step 6: Execution (Run Program)

Execute .exe file by the CPU.

❖ Library Function

A library function in C++ is a pre-written, pre-compiled function that is part of the C++ Standard Library or other external libraries. (ऐसा लाइब्रेरी जिसमें पहले से ही लिखा होता है कि इस function का क्या काम है |)

e.g. = `sqrt()`, `pow()`, `strlen()`,.....

❖ Header File

C++ header files are files with a **.h** extension (old compilers e.g- Turbo C++,..)without .h means **no extension** because use standard library headers like `<iostream>`,`<vector>`,...

(New modern C++) that contain declarations of functions, classes, variables, and other entities. (library function को use करने के लिए हम header file का use करते हैं)

e.g. = `#include <iostream>` , `#include <cmath>`,.....

❖ **Standard Library**

The C++ standard library is a collection of data-structures and algorithms/methods formerly known as the Standard Template Library (STL). which must be provided with any standard-compliant implementation of C++. As a result, using the standard library is portable across different systems and compilers, and does not require downloading and linking additional libraries. It does, however, require the use of header files, as we'll see in a moment. (ये library function की तरह ही होता है जिसके लिए हम `#include <.....>` के लाइन के बाद **using namespace std;** का use करते हैं, जो यह बताता है की जो यह function c++ standard file के अंदर मौजूद है ना की user द्वारा declare किया गया है |)

e.g.

```
#include<iostream>
```

```
using namespace std;
```

❖ **Some Library function & header function with works/purpose**

Header File	Purpose	Example Functions
<iostream>	Input/output streams	cin, cout, endl {not function, its object}
<string>	String handling	length(), substr(), find()
<vector>	Dynamic arrays	push_back(), size(), at()
<algorithm>	Algorithms	sort(), reverse(), find()
<cmath>	Math functions	sqrt(), pow(), sin(), abs()
<cstdlib>	General utilities	rand(), malloc(), exit()
<ctime>	Time manipulation	time(), clock(), difftime()
<map>	Sorted associative containers	insert(), find(), erase()
<set>	Set containers	insert(), find(), count()
<queue>	Queue containers	push(), pop(), front()
<stack>	Stack containers	push(), pop(), top()
<list>	Doubly linked list	push_back(), remove(), sort()
<array>	Fixed-size array	at(), size()
<bitset>	Bit manipulation	set(), reset(), test()
<functional>	Function objects	std::function, bind, mem_fn

Header File	Purpose	Example Functions
<thread>	Multithreading	thread, join, detach
<mutex>	Synchronization primitives	mutex, lock_guard
<chrono>	Time measurement	steady_clock, duration
<fstream>	File handling	ifstream, ofstream, open()
<sstream>	String stream	stringstream, getline()
<numeric>	Numeric operations	accumulate(), iota()
<regex>	Regular expressions	regex_match(), regex_search()

- Cin → take input from user
- Cout → print output
- Endl → change line / new line **OR** \n
- // → Single line comment
- /*.....*/ → Multi line comment
- << → Insertion operator (use in cout)
- >> → Extraction operator (use in cin)

=====HIRA KUMAR=====